GEMSTAR - CAMERA RESEARCH EXTENSION

Task 18.E1 through Task 18.E9

Camera observability, image preprocessing, fixed-point Sobel arithmetic and FPGA preparation.

Status: Software research and hardware architecture preparation completed.

Platform: Windows PC, GEMSTAR, Python and OpenCV.

FPGA hardware measurements: Not performed.

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1. Executive Summary

This extension characterizes GEMSTAR camera recordings and establishes an initial hardware-oriented image-processing reference. Twenty frames from five recordings were extracted and evaluated using image-statistics monitoring, CPU preprocessing benchmarks and integer Sobel arithmetic experiments.

Experiment	Verified output
E3 Camera extraction	20 frames
E5 Observability	8 OBSERVABLE, 12 REVIEW
E6 Timing	2400 timed executions
E7 INT16 Sobel	100% exact software agreement
E8 FPGA preparation	10 planned RTL modules, 12 test cases

2. Dataset and Extraction

Camera SensorView data was verified in five recordings. The task05_fog_test recording had no camera observations in the inspected sample.

Recording	Images	Original resolution
baseline_clear	4	640 x 480
baseline_fog	4	1920 x 1080
baseline_fog_working_radar	4	640 x 480
fog_mor100	4	640 x 480
fog_postfix_validation	4	640 x 480

The baseline_fog images were recorded at 1920 x 1080, while the other extracted images were 640 x 480. For the initial comparative analysis, images were standardized to 640 x 480.

Image-byte counts are consistent with three 8-bit samples per pixel. RGB/BGR channel ordering has not been independently verified.

3. Image Quality Measurements

Recording	Brightness	Contrast	Sharpness	Edges %
baseline_clear	166.64	26.96	529.44	4.74
baseline_fog	209.13	18.21	41.02	1.76
baseline_fog_working_radar	206.82	21.04	96.83	1.55
fog_mor100	189.59	28.11	274.16	3.58
fog_postfix_validation	206.79	21.24	83.63	1.49

The fog-labeled recordings generally showed lower contrast, sharpness or edge density than baseline_clear. However, changes may also reflect scene, rendering and resolution differences. This experiment does not establish a causal effect from fog alone.

4. Camera Observability Monitor

Recording	Observable	Review	Median warnings
baseline_clear	4	0	0.0
baseline_fog	0	4	4.0
baseline_fog_working_radar	0	4	3.0
fog_mor100	4	0	0.0
fog_postfix_validation	0	4	3.0

The monitor used brightness, contrast, Laplacian variance and edge density. Thresholds were derived from four baseline_clear frames and were not calibrated on independent labeled datasets.

OBSERVABLE and REVIEW are provisional statistical states, not validated camera health or obstacle detection correctness labels.

5. CPU Processing Performance

Mode	Median ms	Frame-P95 median ms	Edge F1
GRAYSCALE	0.0501	0.0548	N/A
SOBEL_FULL	0.3149	0.5027	1.0000
GAUSSIAN_SOBEL	0.4039	0.6122	0.5291
SOBEL_HALF	0.1677	0.1915	0.5864

Half-resolution Sobel reduced median execution time compared with full Sobel, but its thresholded output showed substantially less agreement with the full-resolution reference.

All timings are Windows PC/OpenCV measurements. Image loading and initial resolution standardization were outside the timed region. The edge-map F1 is algorithm-to-algorithm agreement, not object detection accuracy.

6. Fixed-Point Sobel Research

Metric	Result
Images evaluated	20
Minimum INT16 agreement	100.000%
Median saturated INT8 agreement	99.039%
Clipped-gradient pixel positions	135690
Maximum observed abs X gradient	630

Metric	Result
Maximum observed abs Y gradient	626

Camera input pixels are unsigned 8-bit. A 3x3 Sobel gradient can theoretically span -1020 to +1020, requiring at least 11 signed bits. The INT8-saturation experiment was deliberately lossy and is not a recommended gradient representation.

Exact integer-gradient agreement was validated in software. This does not verify an actual RTL datapath or its bit-exact rounding, timing, resource use or energy.

7. Proposed Kria FPGA Architecture

The initial accelerator will accept uint8 grayscale pixels, maintain two image row buffers, construct a 3x3 sliding window, compute signed Sobel gradients, combine their absolute magnitudes, and produce an 8-bit output stream.

Hardware property	Planned specification
Input image	640 x 480 grayscale
Input pixel	Unsigned 8-bit
Gradient	Signed 11-bit
Combined magnitude	Widened integer arithmetic
Output	Unsigned 8-bit
Line storage	At least 1280 bytes
Interface	Proposed AXI4-Stream
RTL status	Not implemented

The 10 proposed RTL modules and 12 verification cases are saved in the archive. Task 23 will handle RTL implementation, synthesis, AXI/DMA integration, Kria deployment and hardware measurement.

8. Limitations and Future Work

- Only four selected camera frames per recording.

- No independent camera failure or object labels.

- No verified causal comparison of weather conditions.

- No validated RGB versus BGR image channel ordering.

- No ARM-only versus FPGA hardware comparison.

- No FPGA timing, resources or board energy measured.

- No hardware Sobel RTL implementation completed.

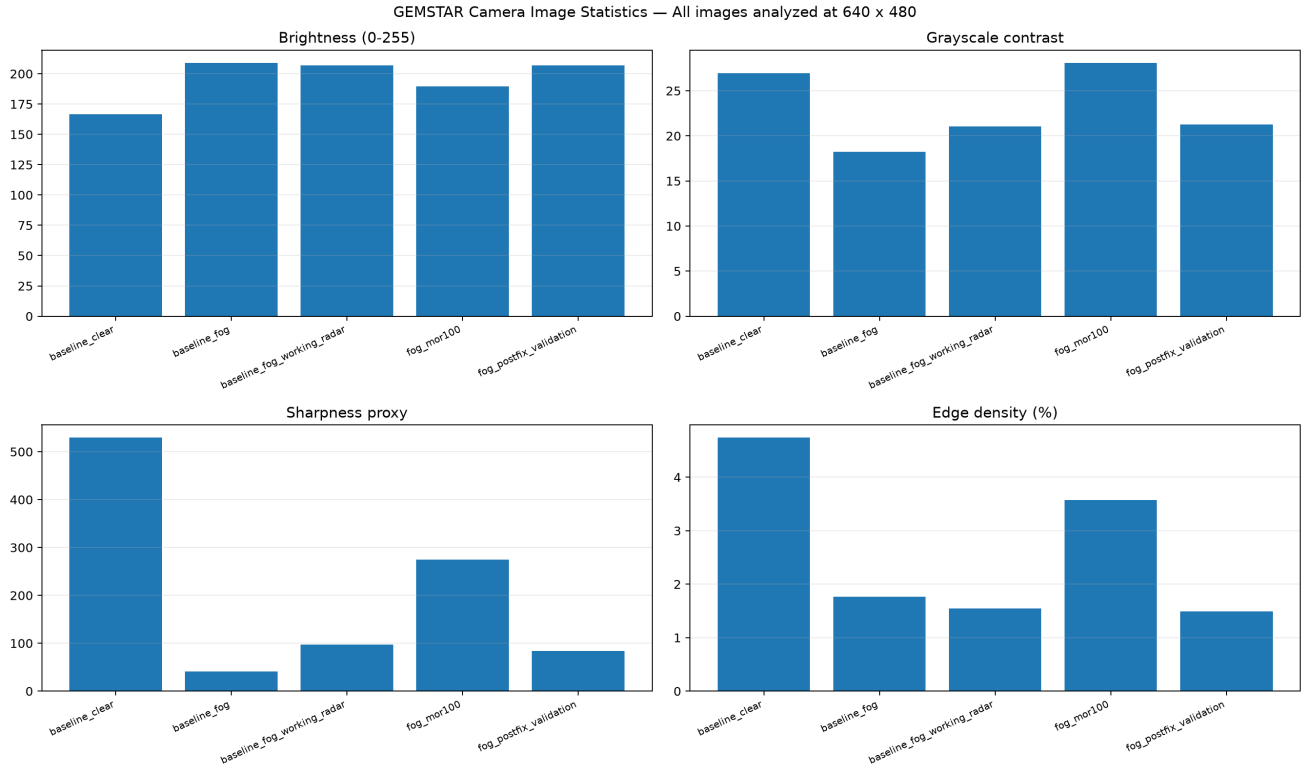
- Additional unseen frames and scenario-matched experiments are needed for robust evaluation.

9. Research Continuation

Task 20 will characterize radar data. Task 21 will investigate synchronized sensor coordinates and timestamps. Task 22 will establish a fusion baseline, and Task 23 will implement selected accelerators on Kria.

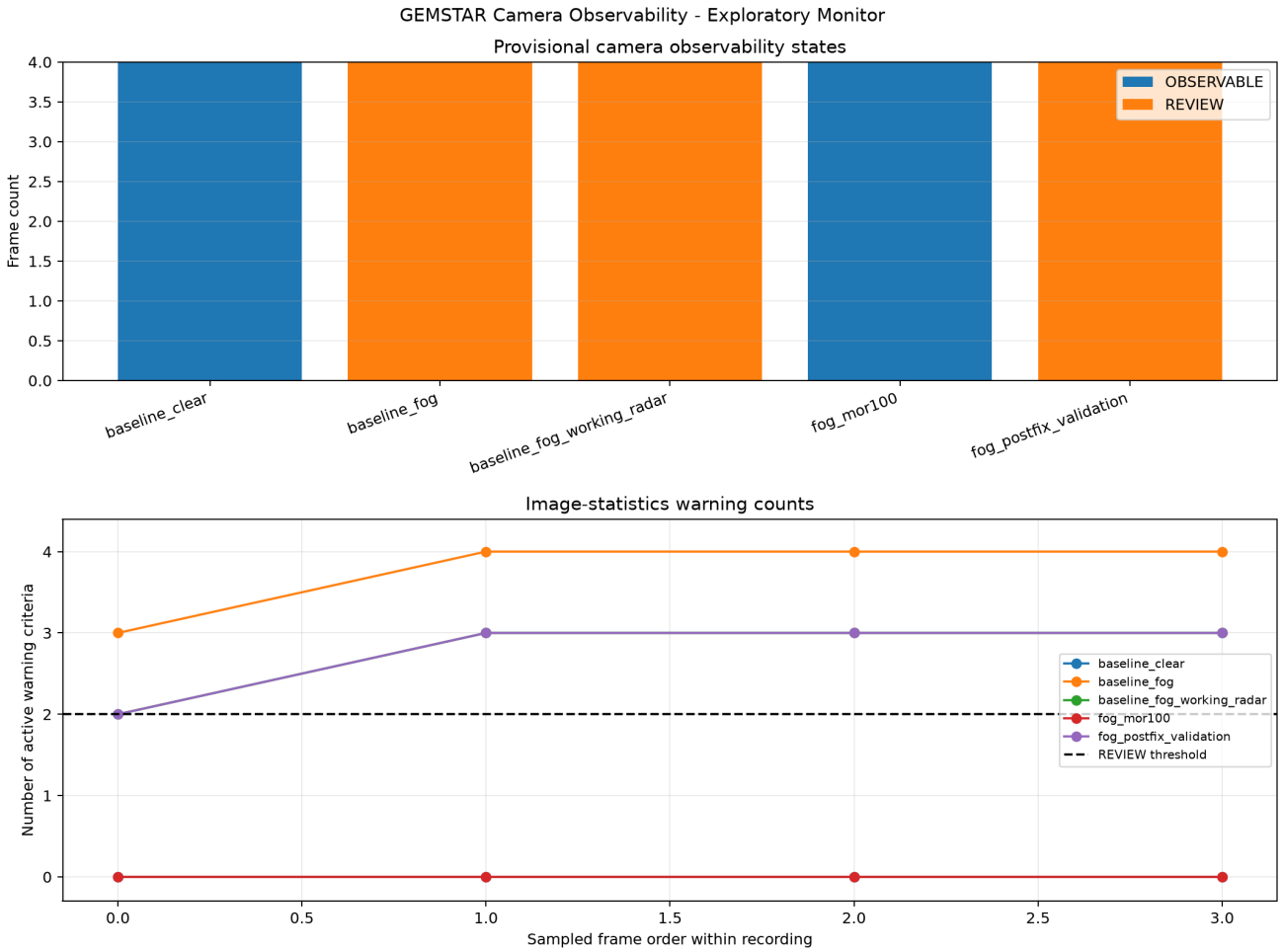
10. Experimental Figures

task18 e4 camera quality comparison



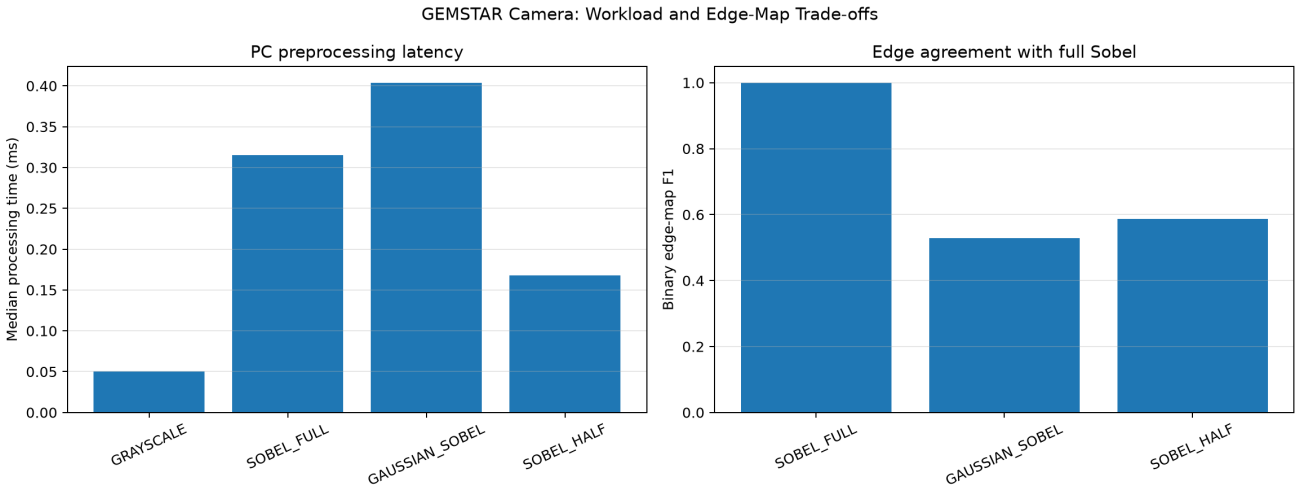
task18_e4_camera_quality_comparison.png

task18 e5 camera observability



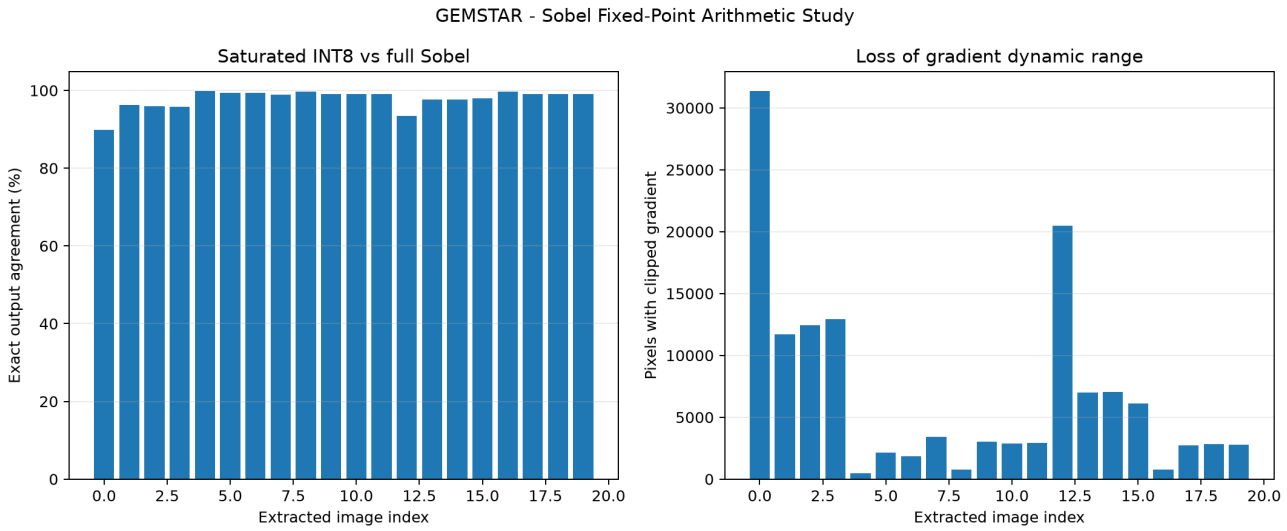
task18_e5_camera_observability.png

task18 e6 camera processing comparison



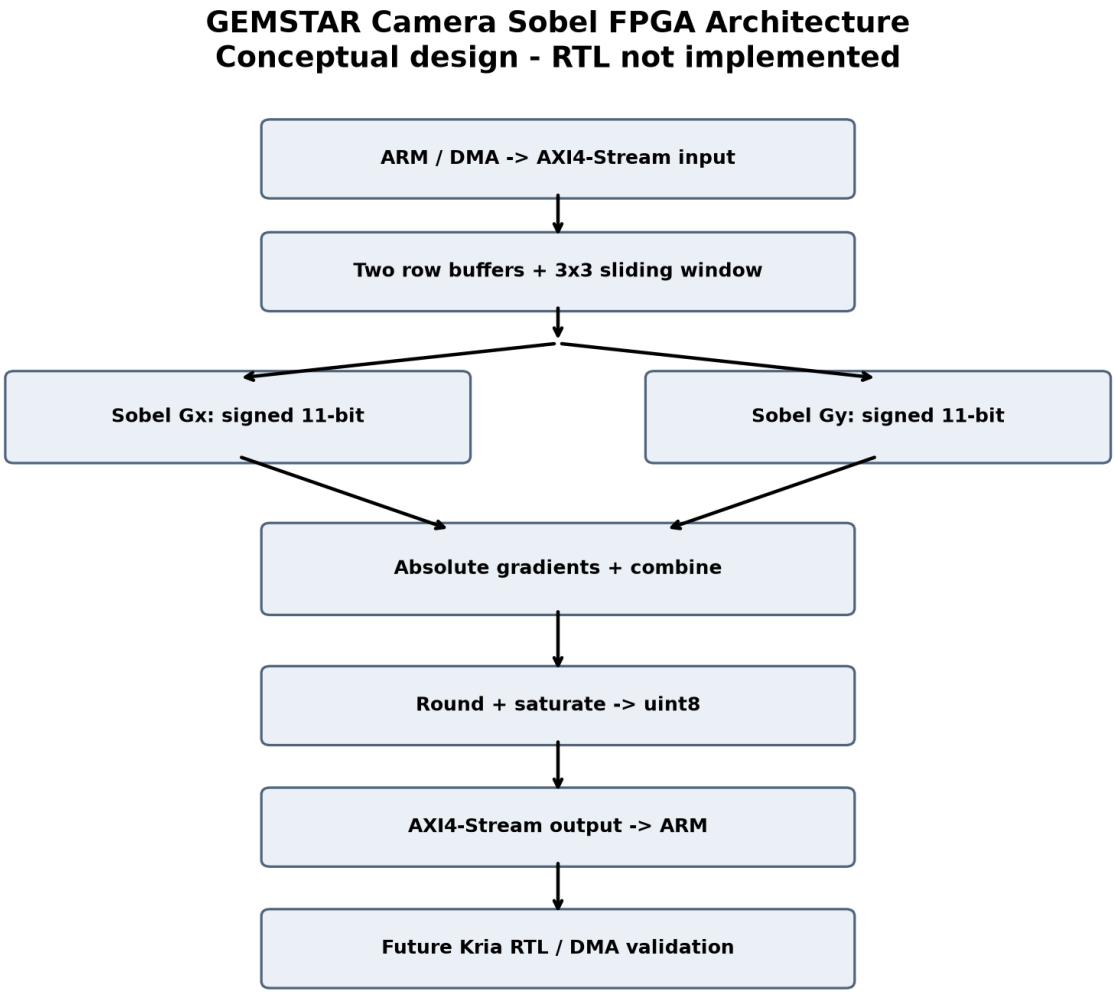
task18_e6_camera_processing_comparison.png

task18 e7 sobel precision comparison



task18_e7_sobel_precision_comparison.png

task18 e8 camera fpga architecture



task18_e8_camera_fpga_architecture.png

11. Extracted GEMSTAR Images

baseline_clear



Original extracted image: baseline_clear_frame_000_rgb.png

baseline_fog



Original extracted image: baseline_fog_frame_000_rgb.png

baseline_fog_working_radar



Original extracted image: baseline_fog_working_radar_frame_000_rgb.png

fog_mor100



Original extracted image: fog_mor100_frame_000_rgb.png

fog_postfix_validation



Original extracted image: fog_postfix_validation_frame_000_rgb.png